$$\frac{\sin\theta_B}{\sin(90^\circ - \theta_B)} = \frac{\mu_2}{\mu_1}$$

or
$$\frac{\sin\theta_B}{\cos\theta_B} = \frac{\mu_2}{\mu_1}$$

$$\therefore \quad \tan\theta_B = \frac{\mu_2}{\mu_1} \qquad \ldots (20.6)$$

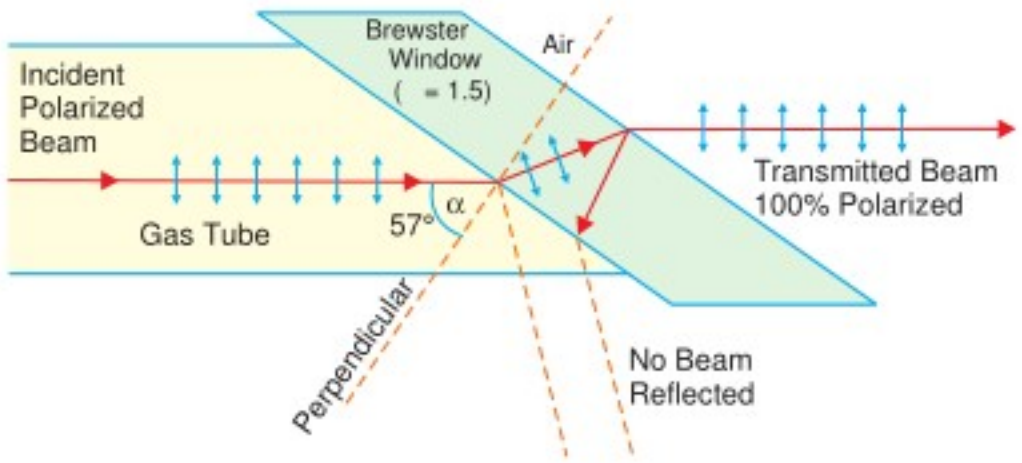
Equ.(20.6) shows that the polarizing angle depends on the refractive index of the reflecting surface. The polarizing angle q_B is known as **Brewster angle**. Light reflected from any angle other than Brewster angle is partially polarized.

Application of Brewster's law:

(*i*) Brewster's law can be used to determine the refractive indices of opaque materials.

(*ii*) It helps us in calculating the polarizing angle necessary for total polarization of reflected light for any material if its refractive index is known. However, the law is not applicable for metallic surfaces.

(*iii*) In gas lasers it is common to arrange two glass windows at the two ends of the laser tube. The windows are arranged at Brewster angle to the axis of the laser tube and hence they are called **Brewster windows**. The light beam traveling between the mirrors of the laser is reflected many times from these mirrors. Since the mirrors are at Brewster angle, all the light that is polarized perpendicular to the beam plane is emitted out of the laser cavity at early stage. In the gas tube remains only radiation polarized in the beam plane. The advantage of this arrangement is that the beam has no reflection losses, since only the transmitted polarized beam is traveling between the mirrors. The radiation out of these lasers is polarized as can be seen in Fig.20.15.



Brewster angle window at the end of gas laser

Fig. 20.15

(*iv*) Another application utilises the Brewster angle for transmitting a light beam into or out of an optical fibre without reflection losses.

20.6.2 POLARIZATION BY REFRACTION - PILE OF PLATES

When unpolarized light is incident at Brewster angle on a smooth glass surface, the reflected light is totally polarized, while the refracted light is partially polarized. If natural light is transmitted through a single plate, the transmitted beam is only partially polarized. If a stack of glass plates is

used instead of a single plate, reflections from successive surfaces occur leading to the filtering of the s-component in the transmitted ray. Ultimately, the transmitted ray consists of p-component alone. It is found that a stack of about 15 glass plates is required for this purpose. The glass plates are supported in a tube of suitable size and inclined at an angle of about 33° to the axis of the tube, as shown in Fig.20.16. Such an arrangement is called a **pile of plates**. Unpolarized light enters the tube and is incident on the plates at Brewster angle and the transmitted light will be totally polarized parallel to the plane of incidence.

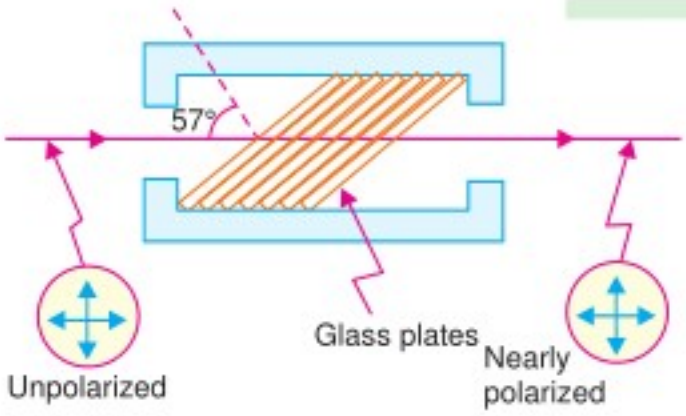


Fig. 20.16

20.6.3 POLARIZATION BY SCATTERING

If a narrow beam of natural light is incident on a transparent medium containing a suspension of ultramicroscopic particles, the light scattered is partially polarized. The incident light causes electrons in the scattering medium to vibrate. A vibrating electron emits most light in a direction perpendicular to its vibration and none along the direction of its vibration. The electric field of the emitted radiation is parallel to the direction of electron vibration. Hence light scattered through about 90° with respect to the incident direction is strongly polarized. The direction of vibration of E vector in the scattered light will be perpendicular to the plane defined by the direction of propagation and the direction of observation, *i.e.*, the plane of the paper, as illustrated in Fig.20.17.

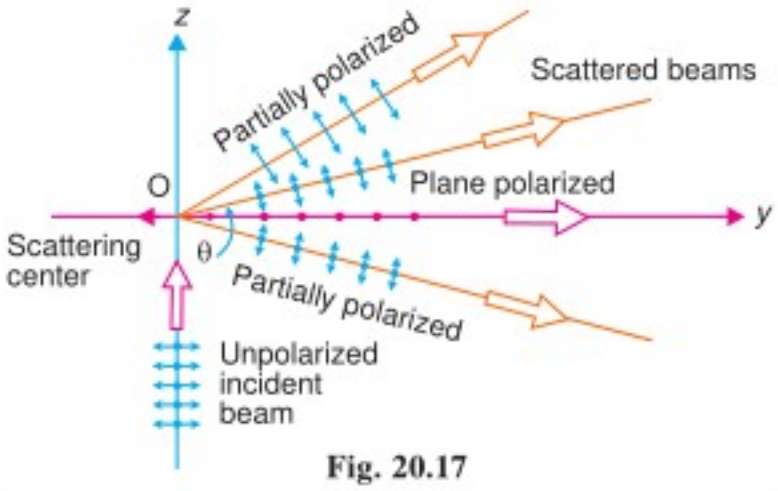


Fig. 20.17

The light from a blue sky is quite strongly polarised, particularly at 90° from the sun. It is not completely polarised because a significant amount of sunlight has undergone multiple-scattering, i.e. has been scattered more than once. Light scattered twice through a total angle of 90° would be less polarised than light scattered once.

20.6.4 POLARIZATION BY SELECTIVE ABSORPTION

A number of crystalline materials absorb more light in one incident plane than another, so that light progressing through the material become more and more polarized as they proceed. This difference in the absorption for the light rays is known as **selective absorption** or **dichroism**. Biot discovered this phenomenon in 1815. When natural light passes through a crystal such as tourmaline, it is split into two components, which are polarized in mutually perpendicular planes. The crystal absorbs light that is polarized in a direction parallel to a particular plane in the crystal but freely transmits the light component polarized in a direction perpendicular to that plane. If the crystal is of proper thickness, one of the components is totally absorbed and the other component emerging from the crystal is